

Notes

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Geometry

Circles

Circumference: $2\pi r$

Surface: πr^2

Cylinders

Surface: $2\pi r^2 + 2\pi r h$

Volume: $V = \pi r^2 h$

Exercises

- Find an expression for the radius r of the base of a cylinder in terms of its volume V given that the ratio of the radius to the height is 3π .

We have $\frac{r}{3\pi} = h$. Let's substitute that in the volume formula:

$$\begin{aligned}\pi r^2 h &= V \\ \pi r^2 \frac{r}{3\pi} &= V \\ \frac{\pi r^3}{3} \pi &= V \\ \frac{r^3}{3} &= V \\ r^3 &= 3V \\ r &= \sqrt[3]{3V}\end{aligned}$$

- Calculate the circumference of the base, given volume is $625\pi m^2$ and the ratio of its height to its radius is 5.

$$\begin{aligned}\pi r^2 h &= V \\ \pi r^2 \cdot (5r) &= 625\pi \\ 5r^3 &= 625 \\ r^3 &= 125 \\ r &= 5\end{aligned}$$

Now the circumference is $2\pi r = 2\pi(5) = 10\pi$

Spheres

Surface: $4\pi r^2$ **Volume:** $\frac{4}{3}\pi r^3$

Triangles

Surface

The formula for the area of a triangle from one angle C and its two adjacent sides a and b is:

$$\frac{1}{2}ab \sin C$$

C is called the **included angle**.

Polygons

The sum of the interior angles of a polygon is always $180 * (n - 2)$ where n is the number of sides.

Regular polygons

With n as the number of sides, exterior angles are given by:

$$\theta = \frac{360}{n}$$

Logarithm

Domain

$$(0, \infty)$$

Range

$$(-\infty, \infty)$$

Polynomials

Forms

- Standard form: $y = ax^2 + bx + c$
- Vertex form: $y = a(x - h)^2 + k$
 - In this form, the coordinates of the vertex are (h, k)
- Factored form: $y = a(x - x_1)(x - x_2)$

Vertex

If the binomial is given in standard form, the vertex' x coordinate is given by $-\frac{b}{2a}$. Once you have the x coordinate, substitute in the equation to find y .

Domain and range

The **domain** is always $(-\infty, +\infty)$.

The **range** is, for upward opening parabolas, assuming vertex form: $y = a(x - h)^2 + k$, the range is $[k, +\infty)$.

Factor theorem

The factor theorem states that $(x - r)$ is a factor of a polynomial $p(x)$ if and only if the root $x = r$ is also a root of $p(x)$.

Exercises

- The polynomial $-5x^4 + 8x^3 + 29x^2 - 20x - 12$ can be expressed as $-(x - 1)(x - 3) \cdot P(x)$. Find $P(x)$.

First let's divide by $(x - 1)$

$$\begin{array}{r|rrrrr} 1 & -5 & 8 & 29 & -20 & -12 \\ & & -5 & 3 & 32 & 12 \\ \hline & -5 & 3 & 32 & 12 & 0 \end{array}$$

So we have $(x - 1) \cdot (-5x^3 + 3x^2 + 32x + 12)$

Let's divide by $(x - 3)$

$$\begin{array}{r|rrrr} 3 & -5 & 3 & 32 & 12 \\ & -15 & -36 & -12 & \\ \hline & -5 & -12 & -4 & 0 \end{array}$$

So we have $(x - 1)(x - 3)(-5x^2 - 12x - 4)$

So we have $(x - 1) \cdot (x - 3) \cdot (-5x^2 - 2x + 4)$

Finally, $-(x - 1)(x - 3)(5x^2 + 2x + 4)$. So $P(x) = 5x^2 + 2x + 4$.

Factoring further, $P(x) = (5x + 2)(x + 2)$.

- Find the polynomial $P(x)$ given that

$$4x^4 - 19x^3 - 9x^2 + 19x + 5 = (x - 1)(x - 5)P(x)$$

First let's divide by $(x - 1)$.

$$\begin{array}{r|rrrrr} 1 & 4 & -19 & -9 & 19 & 5 \\ & & 4 & -15 & -24 & -5 \\ \hline & 4 & -15 & -24 & -5 & 0 \end{array}$$

So $(x - 1)(4x^3 - 15x^2 - 24x - 5)$. Now divide by $(x - 5)$.

$$\begin{array}{r|rrrr} 5 & 4 & -15 & -24 & -5 \\ & & 20 & 25 & 5 \\ \hline & 4 & 5 & 1 & 0 \end{array}$$

So $(x - 1)(x - 5)(4x^2 + 5x + 1)$.

- Find $P(x)$ given $4x^4 - 8x^3 - 3x^2 + 7x - 2 = (x - 2)(x + 1)P(x)$.

First divide by $(x + 1)$:

$$\begin{array}{r|rrrrr} -1 & 4 & -8 & -3 & 7 & -2 \\ & & -4 & 12 & -9 & 2 \\ \hline & 4 & -12 & 9 & -2 & 0 \end{array}$$

So $(x + 1)(4x^3 - 12x^2 + 9x - 2)$. Now divide the remainder by $(x - 2)$:

$$\begin{array}{r|rrrr} 2 & 4 & -12 & 9 & -2 \\ & & 8 & -8 & 2 \\ \hline & 4 & -4 & 1 & 0 \end{array}$$

So $(x + 1)(x - 2)(4x^2 - 4x + 1)$. So $P(x) = (4x^2 - 4x + 1)$. Factoring further,

$$\begin{aligned} P(x) &= 4x^2 - 4x + 1 \\ &= (4x^2 - 2x) + (-2x + 1) \\ &= 2x(2x - 1) - 1(2x - 1) \\ &= (2x - 1)^2 \end{aligned}$$

- Let $p(x) = -6x^4 + 5x^3 + 52x^2 - 19x - 60$. Given that $p(-1) = p(3) = 0$, find the sum of all distinct roots of $p(x)$.

$p(-1) = 0$ means $x = -1$ is a root, so $(x + 1)$ is a factor.

$p(3) = 0$ means $x = 3$ is a root, so $(x - 3)$ is a factor.

Let's first divide by $(x + 1)$.

$$\begin{array}{r|rrrrr} -1 & -6 & 5 & 52 & -19 & -60 \\ & & 6 & -11 & -41 & 60 \\ \hline & -6 & 11 & 41 & -60 & 0 \end{array}$$

So $p(x) = (x + 1)(-6x^3 + 11x^2 + 41x - 60)$. Now let's divide the remainder by $(x - 3)$.

$$\begin{array}{r|rrrr}
 3 & -6 & 11 & 41 & -60 \\
 & & -18 & -21 & 60 \\
 & -6 & -7 & 20 & 0
 \end{array}$$

So $p(x) = (x+1)(x-3)(-6x^2 - 7x + 20)$.

Factoring the remainder further:

$$\begin{aligned}
 p(x) &= (x+1)(x-3)(-6x^2 - 7x + 20) \\
 &= (x+1)(x-3)(-6x^2 + 8x) + (-15x + 20) \\
 &= (x+1)(x-3)(2x+5)(-3x+4)
 \end{aligned}$$

The roots are $x = -1, 3, -\frac{5}{2}, \frac{4}{3}$.

The number we are looking for is $-1 + 3 - \frac{5}{2} + \frac{4}{3} = 2 - \frac{5}{2} + \frac{4}{3} = -\frac{1}{2} + \frac{4}{3} = \frac{5}{6}$.

- Find the distinct roots of $x^4 - 5x^2 + 4$ given $p(1) = p(-1) = 0$.

$(x-1)$ is a factor, let's divide by it first:

$$\begin{array}{r|rrrrr}
 1 & 1 & 0 & -5 & 0 & 4 \\
 & & 1 & 1 & -4 & -4 \\
 & 1 & 1 & -4 & -4 & 0
 \end{array}$$

So $(x-1)(x^3 + x^2 - 4x - 4)$. Now divide by $(x+1)$.

$$\begin{array}{r|rrrr}
 -1 & 1 & 1 & -4 & -4 \\
 & & -1 & 0 & 4 \\
 & 1 & 0 & -4 & 0
 \end{array}$$

So $(x-1)(x+1)(x^2 - 4) = (x-1)(x+1)(x+2)(x-2)$.

- Find $P(x)$ given that $x^4 - 2x^3 - 7x^2 + 18x - 18 = (x-3)(x+3)P(x)$.

First let's divide by $(x-3)$:

$$\begin{array}{r|rrrrr}
 3 & 1 & -2 & -7 & 18 & -18 \\
 & & 3 & 3 & -12 & 18 \\
 & 1 & 1 & -4 & 6 & 0
 \end{array}$$

So the remainder is $x^3 + x^2 - 4x + 6$. Let's divide by $(x+3)$.

$$\begin{array}{r|rrrr}
 -3 & 1 & 1 & -4 & 6 \\
 & & -3 & 6 & -6 \\
 & 1 & -2 & 2 & 0
 \end{array}$$

We have $P(x) = x^2 - 2x + 2$.

- Find $P(x)$ given that $x^4 + x^3 - 36x^2 + 37x - 35 = (x-5)(x+7)P(x)$.

First divide by $(x-5)$:

$$\begin{array}{r|rrrrr}
 5 & 1 & 1 & -36 & 37 & -35 \\
 & & 5 & 30 & -30 & 35 \\
 & 1 & 6 & -6 & 7 & 0
 \end{array}$$

Remainder: $x^3 + 6x^2 - 6x + 7$. Now divide that by $(x+7)$:

$$\begin{array}{r|rrrr}
 -7 & 1 & 6 & -6 & 7 \\
 & & -7 & 7 & -7 \\
 & 1 & -1 & 1 & 0
 \end{array}$$

Result: $x^2 - x + 1$

Exercises

- [illegible]

- What is the remainder when $x^4 - 2x^3 - 4x^2 - 10x - 7$ is divided by $x^2 + 2x + 1$?

[illegible]

- What is the quotient of $2x^4 - x^2 - 1$ divided by $2x^2 + 1$?

$$\begin{array}{r} 2x^2 + 1 \mid \begin{array}{rrrrr} x^2 & -1 & & & \\ 2 & 0 & -1 & 0 & -1 \\ 2 & 0 & 1 & & \\ \hline & & -2 & & \\ & & -2 & 0 & 1 \\ & & & & \hline & & & & 0 \end{array} \end{array}$$

Rational roots theorem

Theorem. Suppose that $\frac{p}{q} \in \mathbb{Q}$, in lowest terms, is a root of a polynomial with integer coefficients. Then p must be a factor of the polynomial's constant term, and q must be a factor of its leading coefficient.

That means every root must take the form $\pm \frac{\text{factor of the constant term}}{\text{factor of the leading coefficient}}$.

Exercises

- For the polynomial $f(x) = 3x^8 - 18x^2 - 33x - 18$, how many potential roots are given by the rational roots theorem?

The numerator must be a factor of -18 . The denominator must be a factor of 3. So here are the possibilities:

$$\pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1, \pm 2, \pm 3, \pm 6, \pm 9, \pm 18$$

So 16.

- Given that the polynomial $2x^3 + x^2 + 5x - 3$ has a root in the interval $(0, \frac{3}{2})$, find the sum of the distinct real roots of $f(x)$.

Using the rational roots theorem, we can find the potential roots of the polynomial. The numerator must be a factor of -3 , and the denominator must be a factor of 2. So here are the possibilities in the interval $(0, \frac{3}{2})$:

$$\frac{1}{2}, 1$$

We can quickly find which one of these roots by evaluating the polynomial at these points. It's $\frac{1}{2}$.

Now let's use synthetic division to find the remaining roots:

$$\begin{array}{r|rrrr} 1/2 & 2 & 1 & 5 & -3 \\ & & 1 & 1 & 3 \\ \hline & 2 & 2 & 6 & 0 \end{array}$$

So $2x^3 + x^2 + 5x - 3 = (x - \frac{1}{2})(2x^2 + 2x + 6)$. Since the discriminant of $2x^2 + 2x + 6$ is negative, it has no real roots. Therefore, the only real root is $x = \frac{1}{2}$.

Rationalizing the denominator

In general, you want to avoid radicals in the denominator. To rationalize the denominator, multiply both the numerator and the denominator by the conjugate of the denominator.

$$\begin{aligned} & \frac{1}{1 + \sqrt{2}} \\ &= \frac{1}{1 + \sqrt{2}} \cdot \frac{1 - \sqrt{2}}{1 - \sqrt{2}} \\ &= \frac{1 - \sqrt{2}}{(1 + \sqrt{2})(1 - \sqrt{2})} \\ &= \frac{1 - \sqrt{2}}{1^2 - \sqrt{2}^2} \\ &= \frac{1 - \sqrt{2}}{1 - 2} \\ &= \frac{1 - \sqrt{2}}{-1} \\ &= -1 + \sqrt{2} \end{aligned}$$

Exercises

-

$$\begin{aligned}
& \frac{1}{1 + \sqrt{2}} \\
&= \frac{1}{1 + \sqrt{2}} \cdot \frac{1 - \sqrt{2}}{1 - \sqrt{2}} \\
&= \frac{1 - \sqrt{2}}{(1 + \sqrt{2})(1 - \sqrt{2})} \\
&= \frac{1 - \sqrt{2}}{1^2 - \sqrt{2}^2} \\
&= \frac{1 - \sqrt{2}}{1 - 2} \\
&= \frac{1 - \sqrt{2}}{-1} \\
&= -1 + \sqrt{2}
\end{aligned}$$

1.

$$\begin{aligned}
& \frac{1 + \sqrt{8}}{\sqrt{9} - \sqrt{2}} \\
&= \frac{(1 + \sqrt{8})(\sqrt{9} + \sqrt{2})}{7} \\
&= \frac{\sqrt{9} + 3\sqrt{8} + \sqrt{2} + \sqrt{16}}{7} \\
&= \frac{3 + 6\sqrt{2} + \sqrt{2} + 4}{7} \\
&= \frac{7 + 7\sqrt{2}}{7} \\
&= 1 + \sqrt{2}
\end{aligned}$$

2.

$$\begin{aligned}
\frac{x^2 y^2}{\sqrt{x^3 y^3}} &= \frac{x^2 y^2 \sqrt{x^3 y^3}}{x^3 y^3} \\
&= \frac{x^2 y^2 x^{3/2} y^{3/2}}{x^3 y^3} \\
&= \frac{x^{7/2} y^{7/2}}{x^3 y^3} \\
&= \sqrt{xy}
\end{aligned}$$

3.

$$\begin{aligned}
\frac{\sqrt{12x}}{3\sqrt{2x^2y}} &= \frac{\sqrt{12x} \cdot \sqrt{2x^2y}}{3 \cdot 2x^2y} \\
&= \frac{\sqrt{24x^3y}}{6x^2y} \\
&= \frac{\sqrt{4x^2 \cdot 6xy}}{6x^2y} \\
&= \frac{2x \cdot \sqrt{6xy}}{6x^2y} \\
&= \frac{\sqrt{6xy}}{3xy}
\end{aligned}$$

4. Given that $d > 0$.

$$\begin{aligned}
\frac{cd}{\sqrt{cd^3}} &= \frac{cd\sqrt{cd^3}}{cd^3} \\
&= \frac{\sqrt{cd^3}}{d^2} \\
&= \frac{\sqrt{c} \cdot d^{3/2}}{d^2} \\
&= \frac{\sqrt{c} \cdot d^{1/2}}{d} \\
&= \frac{\sqrt{cd}}{d}
\end{aligned}$$

5. Given that $p > 0$

$$\begin{aligned}
\frac{7t}{\sqrt{14tp^2}} &= \frac{7t \cdot \sqrt{14tp^2}}{14tp^2} \\
&= \frac{7tp\sqrt{14t}}{14tp^2} \\
&= \frac{\sqrt{14t}}{2p}
\end{aligned}$$

6. Given that $a > 0$

$$\begin{aligned}
\sqrt{\frac{5b}{a^3}} &= \frac{\sqrt{5b}}{\sqrt{a^3}} \\
&= \frac{\sqrt{5b}}{a\sqrt{a}} \\
&= \frac{\sqrt{5ab}}{a\sqrt{a^2}} \\
&= \frac{\sqrt{5ab}}{a^2}
\end{aligned}$$

7.

$$\begin{aligned}
 \sqrt{\frac{14r}{p^5}} &= \frac{\sqrt{14r}}{\sqrt{p^5}} \\
 &= \frac{\sqrt{14p^5r}}{p^5} \\
 &= \frac{\sqrt{14r} \cdot p^{5/2}}{p^5} \\
 &= \frac{\sqrt{14pr}}{p^3}
 \end{aligned}$$

8.

$$\begin{aligned}
 \sqrt{\frac{9a^2}{b^3}} &= \frac{\sqrt{9a^2}}{\sqrt{b^3}} \\
 &= \frac{\sqrt{9a^2}}{b\sqrt{b}} \\
 &= \frac{\sqrt{9a^2b}}{b\sqrt{b}^2} \\
 &= \frac{3a\sqrt{b}}{b^2}
 \end{aligned}$$

9. Given that $z > 0$.

$$\begin{aligned}
 \sqrt{\frac{11}{wz^2}} &= \frac{\sqrt{11}}{\sqrt{wz^2}} \mathfrak{z} = \frac{\sqrt{11}}{z\sqrt{w}} \\
 &= \frac{\sqrt{11w}}{zw}
 \end{aligned}$$

10.

$$\begin{aligned}
 \sqrt{\frac{5a}{3b}} &= \frac{\sqrt{5a}}{\sqrt{3b}} \\
 &= \frac{\sqrt{15ab}}{3b}
 \end{aligned}$$

11.

$$\begin{aligned}
 \sqrt{\frac{a^2+3}{a^2-3}} &= \frac{\sqrt{a^2+3}}{\sqrt{a^2-3}} \\
 &= \frac{\sqrt{(a^2+3)(a^2-3)}}{a^2-3} \\
 &= \frac{\sqrt{a^4-9}}{a^2-3} \\
 &= \frac{\sqrt{a^4-9}}{a^2-3}
 \end{aligned}$$

12. Given $y > 0$

$$\begin{aligned}
 \frac{2 + 3\sqrt{xy}}{3\sqrt{y}} &= \frac{(2 + 3\sqrt{xy}) \cdot \sqrt{y}}{3y} \\
 &= \frac{3y\sqrt{x} + 2\sqrt{y}}{3y} \\
 &= \sqrt{x} + \frac{2\sqrt{y}}{3y}
 \end{aligned}$$

Partial fraction decomposition

We have something of the form:

$$\frac{1}{(x-1)(3x-2)}$$

and we want to decompose it into partial fractions:

$$\frac{1}{(x-1)(3x-2)} = \frac{A}{x-1} + \frac{B}{3x-2}$$

Multiplying both sides by $(x-1)(3x-2)$, we get:

$$1 = A(3x-2) + B(x-1)$$

Setting $x = 1$, we get:

$$1 = A(3(1) - 2) + B(1 - 1) = A$$

Setting $x = \frac{2}{3}$, we get:

$$1 = A\left(3\left(\frac{2}{3}\right) - 2\right) + B\left(\frac{2}{3} - 1\right) = -\frac{B}{3}$$

Solving for B , we get:

$$B = -3$$

So the partial fraction decomposition is:

$$\frac{1}{(x-1)(3x-2)} = \frac{1}{x-1} - \frac{3}{3x-2}$$

Exercises

1. Decompose $\frac{44x + 48}{(x+6)(x-3)(x+2)}$

$$\frac{44x + 48}{(x+6)(x-3)(x+2)} = \frac{A}{x+6} + \frac{B}{x-3} + \frac{C}{x+2}$$

$$44x + 48 = A(x-3)(x+2) + B(x+6)(x+2) + C(x+6)(x-3)$$

Set $x = 3$:

$$44 \cdot 3 + 48 = B \cdot 9 \cdot 5$$

$$\frac{180}{9 \cdot 5} = B$$

$$B = 4$$

Set $x = -2$:

$$\begin{aligned}
 44 \cdot -2 + 48 &= C \cdot 4 \cdot -5 \\
 -40 &= C \cdot -20 \\
 C &= 2
 \end{aligned}$$

Set $x = -6$:

$$\begin{aligned}
 44 \cdot -6 + 48 &= A \cdot -9 \cdot -4 \\
 A &= -6
 \end{aligned}$$

2. Given that

$$\frac{-2x-9}{x+3}x = \frac{1}{x+3} + \frac{B}{x}$$

find B

$$\begin{aligned}
 \frac{-2x-9}{x+3}x &= \frac{1}{x+3} + \frac{B}{x} \\
 (-2x-9) &= x + B(x+3) \\
 -3x-9 &= B(x+3) \\
 \frac{-3(x+3)}{x+3} &= B \\
 B &= -3
 \end{aligned}$$

3. Given that

$$\begin{aligned}
 \frac{-9x+36}{(x-3)(x-6)} &= -\frac{3}{x-3} + \frac{B}{x-6} \\
 -9x+36 &= -3(x-6) + B(x-3) \\
 -6x+18 &= B(x-3) \\
 \frac{-6(x-3)}{x-3} &= B \\
 B &= -6
 \end{aligned}$$

Functions

Horizontal asymptotes of rational functions

Rational functions are of the form:

$$f(x) = \frac{ax^n + bx^{n-1} + \dots}{cx^m + dx^{m-1} + \dots}$$

To get the horizontal asymptotes, you first take the **dominant term**. It's the leading term of the polynomial in the denominator, *without the coefficient*. Here x^m .

Then, you divide every term in both numerator and denominator, and evaluate at $x \rightarrow \infty$.

Radical functions

Domain

All the values where the radicand > 0 for even degrees, $(-\infty, \infty)$ for odd degrees.

Exercises

1. Domain of $d(x) = \sqrt[7]{-2x + \frac{1}{2}} + 1$?

$$-2x + \frac{1}{2} >$$

Roots of rational functions

First factor numerator and denominator, then cancel out common factors, then set the numerator to zero and solve.

Exercises

1. Solve $f(t) = 0$ where $f(t) = \frac{7t^2 + 14}{t^2 - 25}$.

$$\frac{7t^2 + 14}{t^2 - 25} = 0$$

$$\frac{7(t^2 + 2)}{(t + 5)(t - 5)} = 0$$

$$t^2 + 2 = 0$$

$$t^2 = -2$$

2. Solve $f(x) = 0$ where $f(x) = \frac{x^2 + 2x - 15}{x^3 + 5x^2 - 9x - 45}$.

$$\frac{x^2 + 2x - 15}{x^3 + 5x^2 - 9x - 45} = 0$$

$$\frac{x^2 + 5x - 3x - 15}{x^2(x + 5) - 9(x + 5)} = 0$$

$$\frac{(x + 5)(x - 3)}{(x + 5)(x^2 - 9)} = 0$$

$$\frac{(x + 5)(x - 3)}{(x + 5)(x - 3)(x + 3)} = 0$$

$$1 = 0$$

The equation has no solution.

3. Solve $f(t) = 0$ where $f(t) = \frac{t-3}{t^2+6t+5}$

$$\frac{t-3}{t^2+6t+5} = 0$$

$$\frac{t-3}{t^2+t+5t+5} = 0$$

$$\frac{t-3}{(t+5)(t+1)} = 0$$

$$t-3 = 0$$

$$t = 3$$

Invertible functions

A function is invertible on an interval if and only if it is one-to-one on that interval.

To check if a function is one-to-one, we can use the horizontal line test. If any horizontal line intersects the graph of the function at most once, then the function is one-to-one.

Statistics

Variance

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Covariance

Covariance is the measure of the joint variability of two random variables.

$$\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Sum of squares

The sum of squares is the sum of the squared differences between each data point and the mean.

$$\sum_{i=1}^n (x_i - \bar{x})^2$$

Denoted S_{xx} . If two variables are involved (covariance), it is denoted as S_{xy} .

Shortcut formula:

$$S_{xx} = \sum_{i=1}^n x_i^2 - \frac{1}{n \left(\sum_{i=1}^n x_i \right)^2}$$
$$S_{xy} = \sum_{i=1}^n x_i y_i - \frac{1}{n \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}$$

Mean estimation for continuous grouped data

$$\bar{x} \approx \frac{1}{n} \sum_{i=1}^K f_i \cdot m_i$$

where

- f_i is the frequency of the i th interval
- m_i is the midpoint of the i th interval
- n is the total number of observations

Estimating a variance for continuous grouped data

Using the same notation as above:

$$\sigma_n^2 \approx \frac{1}{n} \sum_{i=1}^K f_i \cdot (m_i - \bar{x})^2$$

Graphing

Graphing cubic curves with one distinct real root

If we are given one root of a cubic equation, we can use synthetic division to factor it.

Example

Given that $(x - 3)$ is a factor of $f(x) = x^3 + x^2 - 4x - 24$, sketch the curve of $y = f(x)$.

First, synthetic division:

	x^3	x^2	x^1	x^0
3	1	1	-4	-24
		3	12	24
	1	4	8	0

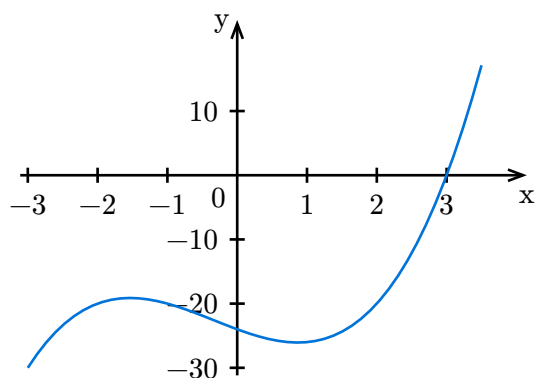
So we have $f(x) = (x - 3)(x^2 + 4x + 8)$

$\therefore x = 3$ is a root.

The discriminant of $x^2 + 4x + 8$ is $4^2 - 4 \cdot 1 \cdot 8 = -16$, which is negative, so $x = 3$ is the only real root.

The y intercept is $f(0) = 0^3 + 0^2 - 4 \cdot 0 - 24 = -24$

Now we can draw the curve:



Logarithms

Domain

$(0, \infty)$

Range

$(-\infty, \infty)$

Complex Numbers

Magnitude

The magnitude of a complex number $z = a + bi$ is defined as $|z| = \sqrt{a^2 + b^2}$.

Argument

For any complex number, like $z = 3 + 4i$, there is $\arg(z)$ that is the angle (in radians) between the x axis and the complex number in the Argand diagram.

So for z :

$$\begin{aligned}\tan \theta &= \frac{4}{3} \\ \theta &= \arctan\left(\frac{4}{3}\right) \\ &\approx 0.93\end{aligned}$$

Trigonometry

Unit circle

Reference angle

The *reference angle* of an angle is the smallest angle between the terminal side of the angle and the x-axis. It is always a positive angle less than or equal to 90 degrees.

Pythagorean identity in the first quadrant

In the first quadrant, any point (x, y) forms a right triangle with the origin, where x and y are the lengths of the legs and $\sqrt{x^2 + y^2}$ is the length of the hypotenuse. Since in the unit circle, the length of the hypotenuse is 1, by definition, the Pythagorean identity states that $x^2 + y^2 = 1$.

Exercises

1. For a point in the first quadrant with $y = \frac{\sqrt{5}}{3}$, find its x coordinate.

$$\sqrt{\left(\frac{\sqrt{5}}{3}\right)^2 + x^2} = 1$$

$$\left(\frac{\sqrt{5}}{3}\right)^2 + x^2 = 1$$

$$x^2 = 1 - \left(\frac{\sqrt{5}}{3}\right)^2$$

$$x^2 = 1 - \frac{5}{9}$$

$$x^2 = \frac{4}{9}$$

$$x = \sqrt{\frac{4}{9}}$$

$$x = \frac{2}{3}$$

2. Point in the first quadrant. y coordinate is $\frac{\sqrt{2}}{3}$. What is the x coordinate?

$$\sqrt{\left(\frac{\sqrt{2}}{3}\right)^2} + x^2 = 1$$

$$\left(\frac{\sqrt{2}}{3}\right)^2 + x^2 = 1$$

$$x^2 = 1 - \left(\frac{\sqrt{2}}{3}\right)^2$$

$$x^2 = 1 - \frac{2}{9}$$

$$x^2 = \frac{7}{9}$$

$$x = \sqrt{\frac{7}{9}}$$

$$x = \frac{\sqrt{7}}{3}$$

Pythagorean trigonometric identity

In the unit circle, since for any central angle $x = \cos \theta$ and $y = \sin \theta$.

That means

$$x^2 + y^2 = 1$$

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

This is the **Pythagorean trigonometric identity**. It works in all quadrants.

Exercises

1. Point P on the unit circle with $x = \frac{2}{5}$. Find $\sin \theta$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\sin \theta = \sqrt{1 - \frac{4}{25}}$$

$$\sin \theta = \sqrt{\frac{21}{25}}$$

$$\sin \theta = \frac{\sqrt{21}}{5}$$

2. y coordinate is $\frac{4}{9}$. Find $\cos \theta$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{4}{9}\right)^2 + \cos^2 \theta = 1$$

$$\cos^2 \theta = 1 - \left(\frac{4}{9}\right)^2$$

$$\cos^2 \theta = 1 - \frac{16}{81}$$

$$\cos^2 \theta = \frac{65}{81}$$

$$\cos \theta = \sqrt{\frac{65}{81}}$$

$$\cos \theta = \frac{\sqrt{65}}{9}$$

3. x coordinate is $\frac{\sqrt{7}}{4}$. Find $\csc \theta$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\sin \theta = \sqrt{1 - \left(\frac{\sqrt{7}}{4}\right)^2}$$

$$\sin \theta = \sqrt{1 - \frac{7}{16}}$$

$$\sin \theta = \sqrt{\frac{9}{16}}$$

$$\sin \theta = \frac{3}{4}$$

$\therefore$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\csc \theta = \frac{4}{3}$$

4. x coordinate is $\frac{3}{7}$. Find $\csc \theta$.

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \frac{9}{49}}$$

$$\sin \theta = \sqrt{\frac{40}{49}}$$

$$\sin \theta = \frac{\sqrt{40}}{7}$$

$\therefore$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\csc \theta = \frac{7}{\sqrt{40}}$$

$$\csc \theta = \frac{7}{2\sqrt{10}}$$

$$\csc \theta = \frac{7}{2\sqrt{10}}$$

$$\csc \theta = \frac{7\sqrt{10}}{20}$$

5. y coordinate is $\frac{1}{9}$. Find $\tan \theta$.

We have $\sin \theta = \frac{1}{9}$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta}$$

$$\cos \theta = \sqrt{1 - \left(\frac{1}{9}\right)^2}$$

$$\cos \theta = \sqrt{1 - \frac{1}{81}}$$

$$\cos \theta = \sqrt{\frac{80}{81}}$$

$$\cos \theta = \frac{\sqrt{80}}{9}$$

$$\cos \theta = \frac{4\sqrt{5}}{9}$$

$\therefore$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\tan \theta = \frac{\frac{1}{9}}{\frac{4\sqrt{5}}{9}}$$

$$\tan \theta = \frac{1}{9} \cdot \frac{9}{4\sqrt{5}}$$

$$\tan \theta = \frac{\sqrt{5}}{20}$$

6. x coordinate is $\frac{\sqrt{11}}{6}$. Find $\cot(\theta)$

We have $\cos \theta = \frac{\sqrt{11}}{6}$ and $\cot(\theta) = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta} = \frac{\frac{\sqrt{11}}{6}}{\sin \theta}$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin^2 \theta = 1 - \left(\frac{\sqrt{11}}{6}\right)^2$$

$$\sin^2 \theta = 1 - \frac{11}{36}$$

$$\sin^2 \theta = \frac{25}{36}$$

$$\sin \theta = \sqrt{\frac{25}{36}}$$

$$\sin \theta = \frac{5}{6}$$

$\therefore$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\cot \theta = \frac{\frac{\sqrt{11}}{6}}{\frac{5}{6}}$$

$$\cot \theta = \frac{\sqrt{11}}{5}$$

Quadrantal angles

Quadrantal angles are angles that are multiples of 90° . For example, 0° , 90° , 180° , 270° , and 360° are all quadrantal angles. They lie on an axis.

They are useful to find the values of trigonometric functions. Just remember that for points on the unit circle, $\cos \theta = x$ and $\sin \theta = y$.

The corresponding angles on $[0, 360^\circ]$ are $0, \frac{\pi}{2}, \pi, 3\frac{\pi}{2}, 2\pi$.

Example: $\cos 0 + \sin\left(\frac{\pi}{2}\right) = 1 + 1 = 2$

Examples

- Find the value of $\tan(180^\circ) + \cot(270^\circ)$

$$\begin{aligned}\tan(180^\circ) + \cot(270^\circ) &= \frac{\sin(180^\circ)}{\cos(180^\circ)} + \frac{\cos(270^\circ)}{\sin(270^\circ)} \\ &= \frac{0}{-1} + \frac{0}{-1} \\ &= 0\end{aligned}$$

- $\sec\left(\frac{\pi}{2}\right)$?

$$\begin{aligned}\sec\left(\frac{\pi}{2}\right) &= \frac{1}{\cos\left(\frac{\pi}{2}\right)} \\ &= \frac{1}{0}\end{aligned}$$

So it's undefined.

Law of sines

In any triangle, where a , b , and c are the lengths of the sides opposite angles A , B , and C respectively, the law of sines states that:

$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

For acute angles, the formula can be flipped:

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$$

For obtuse angles, the same formula works, but it will always give acute angles. To find an obtuse angle, subtract the acute angle from 180° .

Exercises

- Find a if $A = 70^\circ$, $B = 35^\circ$, $b = 12$.

$$\begin{aligned}\frac{a}{\sin(A)} &= \frac{b}{\sin(B)} \\ \frac{a}{\sin(70^\circ)} &= \frac{12}{\sin(35^\circ)} \\ a &= 12 \cdot \frac{\sin(70^\circ)}{\sin(35^\circ)} \\ a &\approx 12 \cdot \frac{0.9397}{0.5736} \\ a &\approx 19.66\end{aligned}$$

- R , Q and P form a triangle. $PR = 21m$. $\angle Q = 89^\circ$, $\angle P = 38^\circ$. What is QR ?

$$\begin{aligned}\frac{QR}{\sin(\angle P)} &= \frac{PR}{\sin(\angle Q)} \\ \frac{QR}{\sin(38^\circ)} &= \frac{21}{\sin(89^\circ)} \\ QR &= \sin(38^\circ) \cdot \frac{21}{\sin(89^\circ)}\end{aligned}$$

- $\triangle XYZ$. $\angle Y = 57^\circ$, $XZ = 35$, $XY = 24$. Find $\angle Z$.

$$\begin{aligned}\frac{\sin(\angle Z)}{XY} &= \frac{\sin(\angle Y)}{XZ} \\ \frac{\sin(\angle Z)}{24} &= \frac{\sin(57^\circ)}{35} \\ \sin(\angle Z) &= 24 \cdot \frac{\sin(57^\circ)}{35} \\ \sin(\angle Z) &\approx 0.575 \\ \angle Z &\approx 35^\circ\end{aligned}$$

Calculus

Limits

Limits of logarithmic functions

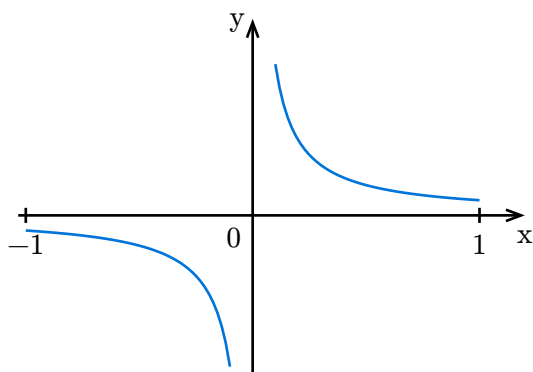
For any real n :

$$\lim_{x \rightarrow n} \log(x) = \log(x)$$

For infinity, remember the domain and apply transformations.

Limits of reciprocal functions

Remember, reciprocal functions have the form $f(x) = \frac{1}{x}$. The limit of a reciprocal function as x approaches infinity is always zero.



In general, for any non-constant polynomial $P(x)$, we have $\lim_{x \rightarrow \pm\infty} \frac{1}{P(x)} = 0$

But for $\lim(x \rightarrow 0)$, we have to consider the degree of the leading term of $P(x)$.

- Any function of the form $y = \frac{1}{x^n}$ where n is odd has the same shape as $y = \frac{1}{x}$. There is no limit at $x = 0$ because left and right limit are not the same.
- Any function of the form $y = \frac{1}{x^n}$ where n is even has the same shape as $y = \frac{1}{x^2}$. There is a limit, since the function will approach ∞ or $-\infty$.

Differentiation

Constant multiple rule

$$\frac{d}{dx}(kf(x)) = k \frac{d}{dx}(f(x))$$

Sum rule

$$\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$$

For example for $y = x^2 + x$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(x^2 + x) \\ &= \frac{d}{dx}(x^2) + \frac{d}{dx}(x) \\ &= 2x + 1 \end{aligned}$$

The power rule

Given a monomial $f(x) = x^n$, where n is a real number,

$$f'(x) = nx^{n-1}$$

Exercises

1. Calculate $\frac{dy}{dx}$ for $y = 2x - \sqrt{x}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(2x - \sqrt{x}) \\ &= \frac{d}{dx}2x - \frac{d}{dx}\sqrt{x} \\ &= 2 - \frac{d}{dx}x^{1/2} \\ &= 2 - \frac{1}{2} \cdot x^{-1/2} \\ &= 2 - \frac{1}{2\sqrt{x}}\end{aligned}$$

2. Find the derivative of $f(x) = \frac{1}{8}x^8$

$$\begin{aligned}f'(x) &= \frac{1}{8} \cdot \frac{d}{dx}x^8 \\ &= \frac{1}{8} \cdot 8x^7 \\ &= \frac{8x^7}{8} \\ &= x^7\end{aligned}$$

3. Derivative of $f(x) = 8\sqrt{x}$

$$\begin{aligned}f'(x) &= \frac{d}{dx}8\sqrt{x} \\ &= \frac{d}{dx}8x^{1/2} \\ &= 8 \cdot \frac{d}{dx}\left(\frac{1}{2} \cdot x^{-1/2}\right) \\ &= 8 \cdot \frac{1}{2} \cdot \frac{1}{2\sqrt{x}} \\ &= \frac{4}{\sqrt{x}}\end{aligned}$$

Rate of change

Average rate of change on an interval $[a, a + h]$:

$$\frac{\delta y}{\delta x} = \frac{f(a + h) - f(a)}{h}$$

The *instantaneous rate of change* at a point $x = a$ is given by:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

Exercises

1. Given that $f(x) = x^2$, what is $f'(2)$ according to the definition?

$$\begin{aligned}
 f'(2) &= \lim_{h \rightarrow 0} \frac{(2+h)^2 - 2^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4 + 4h + h^2 - 4}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4h + h^2}{h} \\
 &= \lim_{h \rightarrow 0} (4 + h) \\
 &= 4
 \end{aligned}$$

Slope of a tangent line

The derivative of a function $f(x)$ at a point $x = a$ is the slope of the tangent line to the graph of $f(x)$ at that point.

So to find the slope of the tangent line at a point (x_1, y_1) , we evaluate $f'(x_1)$.

Exercises

- Find the slope of the tangent to $y = 5x^7 - 3x^5 + 10x^3$ at the point where $x = 1$.

First find $f'(x)$:

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} (5x^7 - 3x^5 + 10x^3) \\
 f'(x) &= 5 \cdot \frac{d}{dx} x^7 - 3 \cdot \frac{d}{dx} x^5 + 10 \cdot \frac{d}{dx} x^3 \\
 f'(x) &= 5 \cdot 7x^6 - 3 \cdot 5x^4 + 10 \cdot 3x^2 \\
 f'(x) &= 35x^6 - 15x^4 + 30x^2 \\
 f'(x) &= 5(7x^6 - 3x^4 + 6x^2)
 \end{aligned}$$

- Find the slope of the tangent to $y = 4x^3 + \frac{5}{2}x^2 - 200x$ at $x = 4$.

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} 4x^3 + \frac{5}{2}x^2 - 200x \\
 &= 12x^2 + 5x - 200
 \end{aligned}$$

The slope is $f'(4) = 12$

- Find the x -coordinate of the point on the curve $y = 2x^2 - 9x - 6$ whose tangent has a slope of 3.

First let's get the derivative of the function:

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} (2x^2 - 9x - 6) \\
 &= 4x - 9
 \end{aligned}$$

Now we set the derivative equal to the slope of the tangent line:

$$\begin{aligned}
 4x - 9 &= 3 \\
 4x &= 12 \\
 x &= 3
 \end{aligned}$$

Integration

The antiderivative

The antiderivative is the opposite of the derivative.

Infinitely many antiderivatives map to the same function.

For example, since the derivative of x^2 is $2x$, we have x^2 as an antiderivative of $2x$. But $x^2 + 2$, $x^2 + 1$, etc. are also antiderivatives of $2x$.

In general, the antiderivatives of $2x$ follow the pattern $x^2 + C$ where C is a constant.

We can also formulate that using the $\int$ symbol for integration:

$$\int 2x \, dx = x^2 + C$$

Where C is called the **constant of integration**.

These integrals are **indefinite integrals**.

The function being integrated (here $2x$) is called the **integrand**.

The power rule for integration

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

Exercises

1. Antiderivative of $\frac{1}{x^4}$?

$$\begin{aligned}\int \frac{1}{x^4} dx &= \int x^{-4} dx \\ &= \frac{x^{-4+1}}{-4+1} + C \\ &= \frac{x^{-3}}{-3} + C \\ &= -\frac{1}{3}x^3 + C\end{aligned}$$

2. Antiderivative of $x^{\frac{5}{8}}$

$$\begin{aligned}\int x^{\frac{5}{8}} dx &= \frac{x^{\frac{5}{8}+1}}{\frac{5}{8}+1} + C \\ &= \frac{x^{\frac{13}{8}}}{\frac{13}{8}} + C \\ &= \frac{8x^{\frac{13}{8}}}{13} + C\end{aligned}$$

3. Antiderivative of $\sqrt[5]{x^7}$

$$\begin{aligned}\int x^{\frac{7}{5}} dx &= \frac{x^{7/5+1}}{7/5+1} + C \\ &= \frac{x^{12/5}}{12/5} \\ &= \frac{5x^{12/5}}{12}\end{aligned}$$

4. Calculate $\int \left(\frac{z}{5} \, dz\right)$

$$\begin{aligned}
 \int \left(\frac{z}{5} dz \right) &= \frac{1}{5} \int z \, dz \\
 &= \frac{1}{5} \cdot \left(\frac{z^2}{2} + C \right) \\
 &= \frac{z^2}{10} + C
 \end{aligned}$$

The constant factor rule for indefinite integrals

To take the integral of a power function with a constant factor, like $5x^2$, we can take the constant factor out of the integral:

$$\int 5x^2 dx = 5 \cdot \int x^2 dx$$

This is the **constant factor rule**. In general:

$$\int k f(x) dx = k \int f(x) dx$$

Where k is a **constant**. It does not work if k is a variable.

Exercises

1. Calculate $\int \frac{1}{2y^5} dy$

$$\begin{aligned}
 \int \frac{1}{2y^5} dy &= \frac{1}{2} \int y^{-5} \\
 &= \frac{1}{2} \cdot \frac{y^{-5+1}}{-5+1} + C \\
 &= \frac{1}{2} \cdot \frac{y^{-4}}{-4} + C \\
 &= \frac{1}{2} \cdot -\frac{1}{4y^4} + C \\
 &= -\frac{1}{8y^4} + C
 \end{aligned}$$

2. Calculate $\int \frac{1}{3x^6} dx$

$$\begin{aligned}
 \int \frac{1}{3x^6} dx &= \frac{1}{3} \cdot \int x^{-6} \\
 &= \frac{1}{3} \cdot \frac{x^{-5}}{-5} + C \\
 &= \frac{1}{3} \cdot -\frac{1}{5x^5} + C \\
 &= -\frac{1}{15x^5} + C
 \end{aligned}$$

3. Calculate $\int \frac{\pi}{2} dz$

$$\begin{aligned}
 \int \frac{\pi}{2} dz &= \frac{\pi}{2} \int z^0 dz \\
 &= \frac{\pi}{2} \cdot z + C \\
 &= \frac{\pi z}{2} + C
 \end{aligned}$$

The sum rule for indefinite integrals

To take the integral of a sum, you can take the sum of the integrals of each term:

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

The sum rule can be combined with the constant factor rule to integrate sums of constant multiples of functions:

$$\int (k_1 f(x) + k_2 g(x)) dx = k_1 \int f(x) dx + k_2 \int g(x) dx$$

Exercises

- Calculate $\int \sqrt{x}(2+x) dx$

$$\begin{aligned}
 \int \sqrt{x}(2+x) dx &= \int 2\sqrt{x} + x \cdot \sqrt{x} dx \\
 &= \int 2\sqrt{x} dx + \int x \cdot \sqrt{x} dx \\
 &= 2 \int \sqrt{x} dx + \int x^{3/2} dx \\
 &= 2 \cdot \frac{x^{3/2}}{\frac{3}{2}} + \frac{x^{5/2}}{\frac{5}{2}} + C \\
 &= \frac{4x^{3/2}}{3} + \frac{2x^{5/2}}{5} + C \\
 &= \frac{4\sqrt{x^3}}{3} + \frac{2\sqrt{x^5}}{5} + C
 \end{aligned}$$

- Calculate $\int \frac{2+x^4}{x^2} dx$

$$\begin{aligned}
 \int \frac{2+x^4}{x^2} dx &= \int \frac{2}{x^2} dx + \int \frac{x^4}{x^2} dx \\
 &= \int 2 \cdot x^{-2} dx + \int x^2 dx \\
 &= 2 \cdot \int x^{-2} dx + \frac{x^3}{3} + C \\
 &= 2 \cdot -x^{-1} + \frac{x^3}{3} + C
 \end{aligned}$$

- Calculate:

$$\begin{aligned}
\int (3 - x^2)^2 dx &= \int (9 - 6x^2 + x^4) dx \\
&= 9 \int 1 dx - 6 \int x^2 dx + \int x^4 dx \\
&= 9x - 6 \frac{x^3}{3} + \frac{x^5}{5} + C \\
&= \frac{x^5}{5} - 2x^3 + 9x + C
\end{aligned}$$

- Find the antiderivative of $z^3 + 2z - 1$

$$\begin{aligned}
\int z^3 + 2z - 1 dz &= \int z^3 dz + \int 2z dz - \int 1 dz \\
&= \frac{z^4}{4} + 2 \frac{z^2}{2} - z + C \\
&= \frac{z^4}{4} + z^2 - z + C
\end{aligned}$$